

Excercise 3 and 4: Assertion And AAA

AssertionTest.java

```
1 package org.example;
2
3 import org.junit.Test;
4
5 import static org.junit.Assert.*;
6
7 public class AssertionsTest {
8
9     @Test
10    public void testAssertions() {
11
12        // Assert Equals
13        assertEquals( expected: 5, actual: 2 + 3);
14
15        // Assert True
16        assertTrue( condition: 5 > 3);
17
18        // Assert False
19        assertFalse( condition: 5 < 3);
20
21        // Assert Null
22        assertNull( object: null);
23
24        // Assert Not Null
25        assertNotNull(new Object());
26    }
27 }
```

output

```
Run  AssertionsTest x
[Icons]
v ✓ AssertionsTest (org.example) 3 ms ✓ 1 te
  ✓ testAssertions 3 ms "C:\
Proc
```

CalculatorAAATest.java

```
import org.junit.After;
import org.junit.Before;
import org.junit.Test;

import static org.junit.Assert.*;

public class CalculatorAAATest {

    private Calculator calculator; 3 usages

    @Before
    public void setup() {
        calculator = new Calculator();
        System.out.println("Setup Completed");
    }

    @After
    public void teardown() { System.out.println("Teardown Completed"); }

    @Test
    public void testAddition() {

        // Arrange
        int a = 10;
        int b = 20;

        // Act
        int result = calculator.add(a, b);
```

```
        // Act
        int result = calculator.multiply(a, b);

        // Assert
        assertEquals("expected: 20, result);

    }
}
```

Run CalculatorAAATest x

✓ 2 tests passed 2 tests total, 8 ms

✓ testMultiplication	8 ms
✓ testAddition	0 ms

"C:\Program Files\Java\jdk-21\bin\
Setup Completed
Teardown Completed
Setup Completed
Teardown Completed
Process finished with exit code 0